



**BPM 1
CPR**

**IMCQ 3
Fall 2025**

KEY

1. A 19-year-old man comes to the emergency department because of a 2-hour history of severe epigastric pain radiating to his back. He has had several similar but milder episodes in the past. He denies alcohol use. Physical examination shows tenderness in the epigastric region without rebound or guarding. Laboratory serum studies show:

Serum amylase: Elevated

Serum lipase: Elevated

Serum triglycerides: 850 mg/dL (Normal: <150 mg/dL)

Total cholesterol: 150 mg/dL (normal: <200 mg/dL)

Which of the following apolipoproteins is most likely deficient in this patient?

- | | |
|-------------------------|-----|
| A. Apolipoprotein A-I | 14% |
| B. Apolipoprotein B-100 | 14% |
| C. Apolipoprotein C-II | 14% |
| D. Apolipoprotein D | 14% |
| E. Apolipoprotein E | 14% |
| F. Apolipoprotein B48 | 14% |
| G. Apolipoprotein (a) | 14% |

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- D. Apolipoprotein D
- E. Apolipoprotein E
- F. Apolipoprotein B48
- G. Apolipoprotein (a)

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0223; Discuss the functions and effects of deficiency of apo B-48, apo B-100, apo C-II, apo E.

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0231; Differentiate different hyperlipidemias (Type I, IIa, IIb, III, IV and V) using lipid profile (Serum TAG and Serum cholesterol), lipoprotein electrophoresis, clinical features and molecular mechanisms and inheritance patterns.

2. A 54-year-old man comes to the physician for a routine health maintenance examination. At his previous visit 1 year ago, he was advised to exercise regularly, stop smoking, and increase dietary fiber intake to reduce his cardiovascular risk. Since that time, he has lost 2.2 kg (5 lb). He is 170.2 cm (5 ft 7 in) tall and weighs 90.7 kg (200 lb); BMI is 31.3 kg/m². The physician recommends adopting a Mediterranean diet and increasing physical activity. Which of the following findings would most likely indicate an increased risk of cardiovascular disease in this patient?

- | | |
|--|----|
| A. Serum LDL-cholesterol concentration of 80 mg/dL (N:<160) | 0% |
| B. Blood pressure of 120/80 mmHg | 0% |
| C. Waist circumference of 85 cm (33.5 in) | 0% |
| D. Serum total cholesterol concentration of 180 mg/dL (N:<200) | 0% |
| E. Serum HDL-cholesterol concentration of 5 mg/dL (N:40-60) | 0% |
| F. Higher visceral fat compared to subcutaneous fat | 0% |
| G. Increased sensitivity of tissues to insulin | 0% |
| H. Low plasma glucose levels | 0% |
| I. Low serum homocysteine levels | 0% |
| J. Increased endothelial lipoprotein lipase activity | 0% |

2. A 54-year-old man comes to the physician for a routine health maintenance examination. At his previous visit 1 year ago, he was advised to exercise regularly, stop smoking, and increase dietary fiber intake to reduce his cardiovascular risk. Since that time, he has lost 2.2 kg (5 lb). He is 170.2 cm (5 ft 7 in) tall and weighs 90.7 kg (200 lb); BMI is 31.3 kg/m². The physician recommends adopting a Mediterranean diet and increasing physical activity. Which of the following findings would most likely indicate an increased risk of cardiovascular disease in this patient?

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SOM.MK.III.BPM1.1.CPR.2.BCHM. 0242

Highlight role of protective factors against development of coronary heart disease (CHD) like fiber, moderate alcohol, and vitamins that decrease homocysteine levels.

SOM.MK.III.BPM1.1.CPR.2.BCHM. 0238

List the diagnostic criteria for obesity: BMI, waist circumference, waist/hip ratio

SOM.MK.III.BPM1.1.CPR.2.BCHM. 0236

Describe risk factors for CHD related to abnormal lipoprotein pattern.

3. A 15-month-old boy is brought to the physician by his parents because of a 3-month history of foul-smelling, greasy stools. He has a history of delayed developmental milestones, failure to thrive and difficulty seeing at night. He is at the 5th percentile for height and weight. Physical examination shows a malnourished child. Laboratory studies show anemia with abnormally shaped red blood cells, low serum cholesterol and triglycerides. A rare inherited disorder is suspected by the physician. Which of the following processes is most likely defective in this patient?

- | | |
|---|----|
| A. Clearance of chylomicron remnants and IDL | 0% |
| B. Microsomal triglyceride transfer protein assembly of lipoproteins | 0% |
| C. Cellular uptake of low-density lipoprotein | 0% |
| D. Hydrolysis of triglycerides in circulating chylomicrons | 0% |
| E. Reverse cholesterol transport | 0% |
| F. Transfer of cholesterol ester and triglycerides between lipoproteins | 0% |

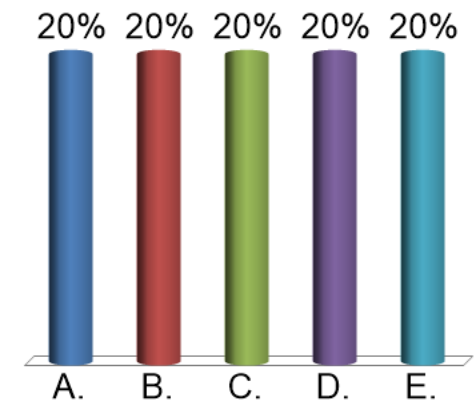
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SOM.MK.I.BPM1.1.CPR.2.BCHM. 0229; Describe hypoalphalipoproteinemia (Tangier disease) and abetalipoproteinemia and hypobetalipoproteinemia (molecular mechanism, lipid profile abnormalities, basis for clinical features)

4. A 25-year-old man is brought to the emergency department 30 minutes after he was involved in a motor vehicle collision. He is bleeding and has lost a significant amount of blood from the wounds on his extremities. His pulse is 130/min and blood pressure is 80/50 mmHg. Physical examination shows pallor, diaphoresis, and increased capillary refill time. Central venous pressure and cardiac output are low. Which of the following alterations in the vascular function curve is most likely in this patient?

- A. Downward and leftward shift
- B. Upward and rightward shift
- C. Decreased slope without a change in the mean filling pressure
- D. Increased slope without a change in the mean filling pressure
- E. No change in the curve



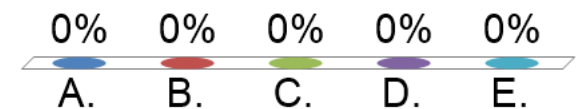
4. A 25-year-old man is brought to the emergency department 30 minutes after he was involved in a motor vehicle collision. He is bleeding and has lost a significant amount of blood from the wounds on his extremities. His pulse is 130/min and blood pressure is 80/50 mmHg. Physical examination shows pallor, diaphoresis, and increased capillary refill time. Central venous pressure and cardiac output are low. Which of the following alterations in the vascular function curve is most likely in this patient?

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- E. No change in the curve

SOM.MK.II.BPM1.3.CPR.2.PHYS.0141 Explain the vascular function curve and how changes in total peripheral resistance, blood volume, and venous compliance influence this curve
Downward and leftward shift of the curve due to reduced mean systemic filling pressure

5. A 30-year-old healthy woman participates in a study related to the changes in cardiovascular parameters during exercise. She is instructed to run 3 miles at a slow pace but without a break. Her venous return, cardiac output, and right atrial pressure are recorded at rest and end of the run, and cardiac and vascular function curves are constructed. The points of intersection of these curves at rest and after exercise are compared. Compared to rest, which of the following alterations in the point of intersection of these curves is most likely at the end of the run in this participant?

- A. Shifts downward and to the right
- B. Shifts downward and to the left
- C. Shifts upward and to the left
- D. Shifts upward and to the right
- E. No change in the position of the point



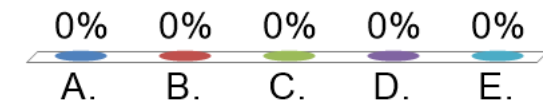
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- C. Shifts upward and to the left
- ✓ D. Shifts upward and to the right
- E. No change in the position of the point

SOM.MK.II.BPM1.3.CPR.2.PHYS.0143 Explain how the intersection point of the cardiac and vascular function curves changes in certain situations such as exercise, use of vasopressor drugs, fluid infusion, acute hemorrhage, heart failure, and use of positive inotropic agents like digoxin.

6. A 68-year-old man comes to the clinic because of a 3-week history of recurrent episodes of palpitations and dizziness. He is diagnosed with a supraventricular tachycardia (SVT) and prescribed verapamil, a calcium channel blocker. His condition improves after this treatment but his resting heart rate decreases. Which of the following mechanisms best explains the reduction in heart rate observed in this patient after treatment?

- A. Blockage of L-type Ca^{2+} channels in the pacemaker cells (phase 0)
- B. Blockage of T-type Ca^{2+} channels in ventricular cells (phase 4)
- C. Blockage of fast Na^{+} channels in ventricular cells (phase 0)
- D. Prolongation of phase 2 (plateau) of the ventricular cells
- E. Enhances K^{+} efflux during phase 3 of the pacemaker cells



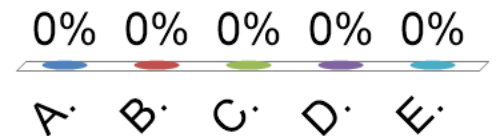
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- E. Enhances K^{+} efflux during phase 3 of the pacemaker cells

SOM.MK.II.BPM1.3.CPR.2.PHYS.0164 Compare and contrast the ventricular myocyte action potential and the pacemaker action potential in terms of their phases and ionic basis. Verapamil blocks L-type Ca^{2+} channels, which are critical for phase 0 depolarization in pacemaker cells but not in ventricular myocytes

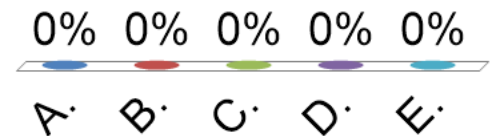
7. A 25-year-old woman comes to the physician for a routine examination. She is a trained athlete who practices endurance exercise. She has no major illness. Her pulse is 52/min; other vital signs are within normal limits. Physical examination shows no abnormalities. Results of laboratory studies are within the reference ranges. An ECG shows sinus bradycardia. The physician says that her bradycardia is because of an increased vagal tone. Increased vagal tone reduces the slope of the phase 4 of the SA nodal action potential. Which of the following ionic mechanisms is most likely responsible for this phase of the action potential?

- A. K^+ efflux
- B. Ca^{2+} efflux
- C. Inward Na^+ current
- D. Na^+ influx through voltage-gated sodium channels
- E. K^+ influx



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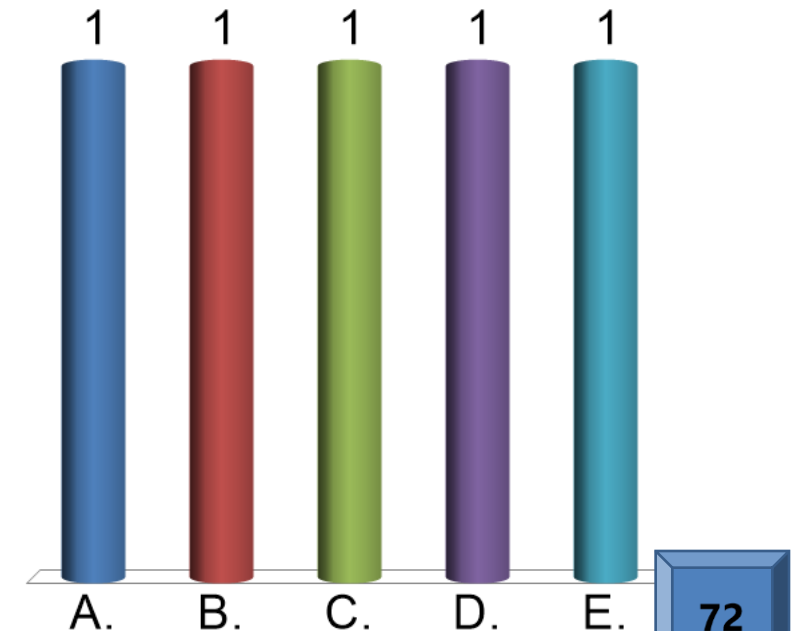
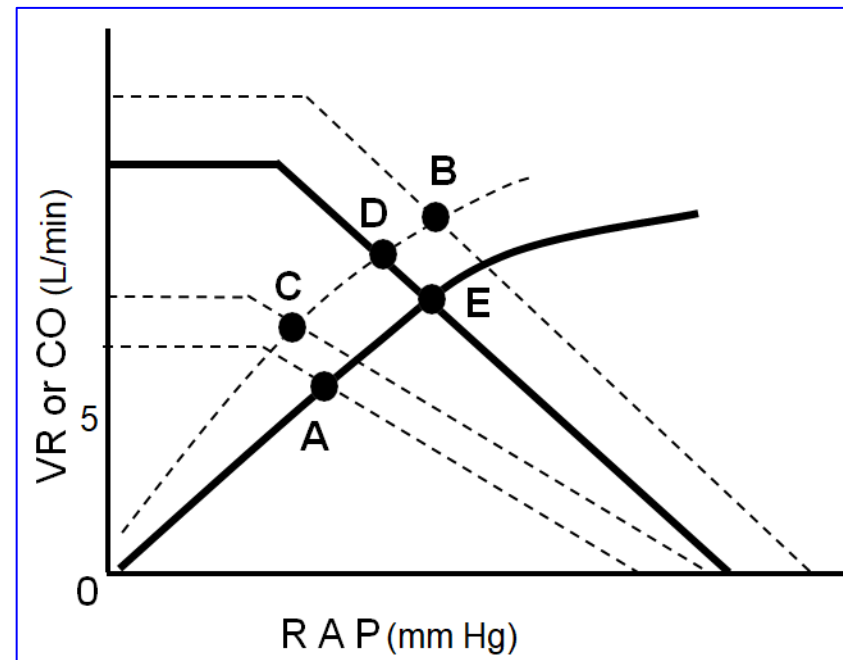
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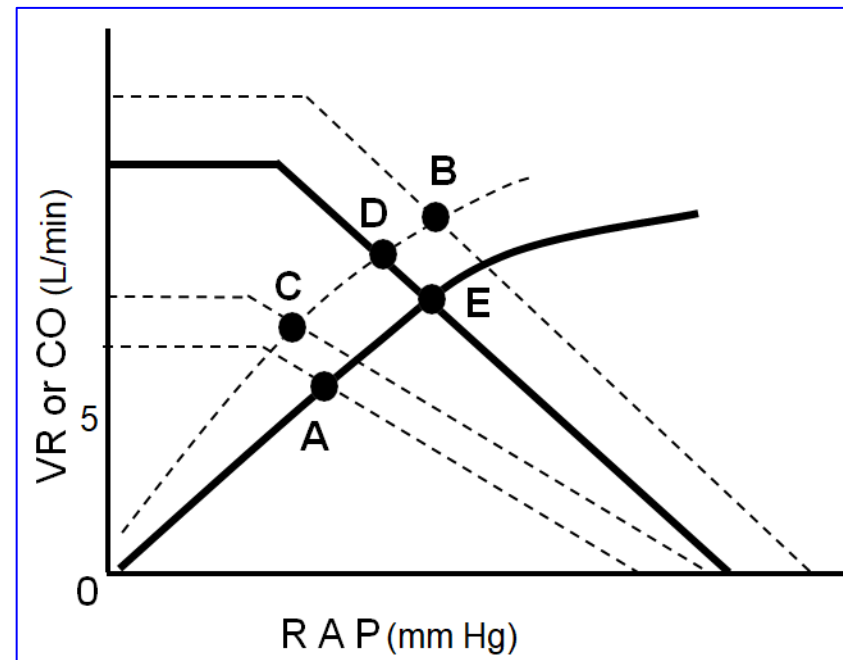
8. A 31-year-old woman is brought to the emergency department 2 hours after she was involved in a motor vehicle collision. She sustained open wounds and had a mild blood loss. Her pulse is 122/min, respirations are 20/min, and blood pressure is 110/74 mm Hg. She appears anxious. Physical examination shows cold and sweaty skin, and a low volume pulse. The vascular function curve and cardiac function curve before bleeding are shown with solid black curves in the graph. Which of the following points of intersections in the graph best represents this patient's steady state at the time of arrival in the hospital?

- A. A
- B. B
- C. C
- D. D
- E. E



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9. A 31-year-old woman comes to the physician because of a 2-week history of fatigue and occasional palpitations. She has a family history of heart disease. Vital signs are within normal limits. Physical examination shows no abnormalities. An ECG in lead II is shown in the figure. Which of the following values best approximates the QT interval in this patient? (Each small box=0.04 sec)



- A. 0.24 second
- B. 0.34 second
- C. 0.40 second
- D. 0.56 second
- E. 0.84 second

QT = 10 small boxes = $0.04 \times 10 = 0.40$ second

0 0 0 0 0
A. B. C. D. E.



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SOM.MK.II.BPM1.3.CPR.2.PHYS.0172 Name the parts of a typical bipolar (Lead II) ECG tracing and explain the relationship between each of the waves, intervals, and segments in relation to the electrical state of the heart



10. A 48-year-old woman comes to the physician for a follow-up examination. She was diagnosed with mitral stenosis 2 years ago. She says she has shortness of breath for the past 3 days. Her pulse is 88/min, respirations are 18/min, and blood pressure is 122/74 mm Hg. Physical examination shows an irregularly irregular pulse and a loud second heart sound in the pulmonary area. An ECG shows absent P waves, changing R-R intervals, and narrow QRS complex. She is diagnosed with atrial fibrillation as a complication of mitral stenosis. Which of the following electrical events is most likely absent in this patient's heart?

- A. Atrial depolarization
- B. Conduction through the AV node
- C. Spread through the Purkinje system
- D. Ventricular depolarization
- E. Ventricular repolarization
- F. Conduction through the bundle of His

0
0
0
0
0
0



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0
0
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0
0



11. A 72-year-old man comes to the physician because of a 5-week history of easy fatigability. His blood pressure is 124/82 mm Hg. Physical examination shows no abnormalities. An ECG in lead II is shown below. He is diagnosed with first-degree heart block. Which of the following values best approximates the heart rate of this patient?



- A. 50/min
- B. 62/min
- C. 75/min
- D. 84/min
- E. 96/min

0 0 0 0 0
A B C D E



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- A. 50/min
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- ✓ C. 75/min
- D. 84/min
- E. 96/min

Heart rate= $300/\text{Large boxes}$
 $=300/4 = 75/\text{min}$

SOM.MK.II.BPM1.3.CPR.2.PHYS.0172 Name the parts of a typical bipolar (Lead II) ECG tracing and explain the relationship between each of the waves, intervals, and segments in relation to the electrical state of the heart



12. A 62-year-old woman comes to the physician for a follow-up examination. She has an 8-year history of hypertension, and she is not adherent to anti-hypertensive medications. Her blood pressure is 146/88 mm Hg. ECG recorded in aVF shows equiphasic QRS complex. ECG recorded in Lead I is shown. Which of the following values best approximates the mean electrical axis (MEA) of this patient's heart?



Lead I

- A. +Zero degree
- B. +30 degree
- C. +90 degree
- D. -120 degree
- E. -30 degree
- F. -60 degree

0 0 0 0 0 0
A B C D E F

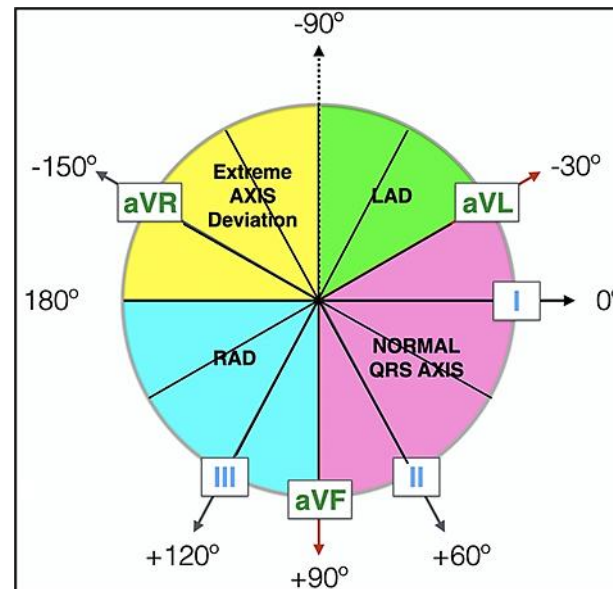


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- D. -120 degree
- E. -30 degree
- F. -60 degree



SOM.MK.II.BPM1.3.CPR.2.PHYS.0177 Discuss the concept of mean electrical axis of the heart, its normal range and the different methods to determine it.



13. A 68-year-old man comes to the physician because of a 3-week history of two episodes of chest pain. He says the pain occurred when he was lifting heavy weights, lasted for 3 minutes, and is relieved by rest. He has a 15-year history of type 2 diabetes mellitus. His blood pressure is 154/90 mm Hg. Physical examination shows a loud S2 in the aortic area. He is asked to perform a treadmill stress test. An ECG recorded during this test is shown. Which of the following ECG abnormalities is most likely in this patient?



- A. Inverted P waves
- B. Tall T waves
- C. Equiphasic QRS waves
- D. ST segment depression
- E. ST segment elevation

0	0	0	0	0
☐	☐	☐	☒	☐
A.	B.	C.	D.	E.

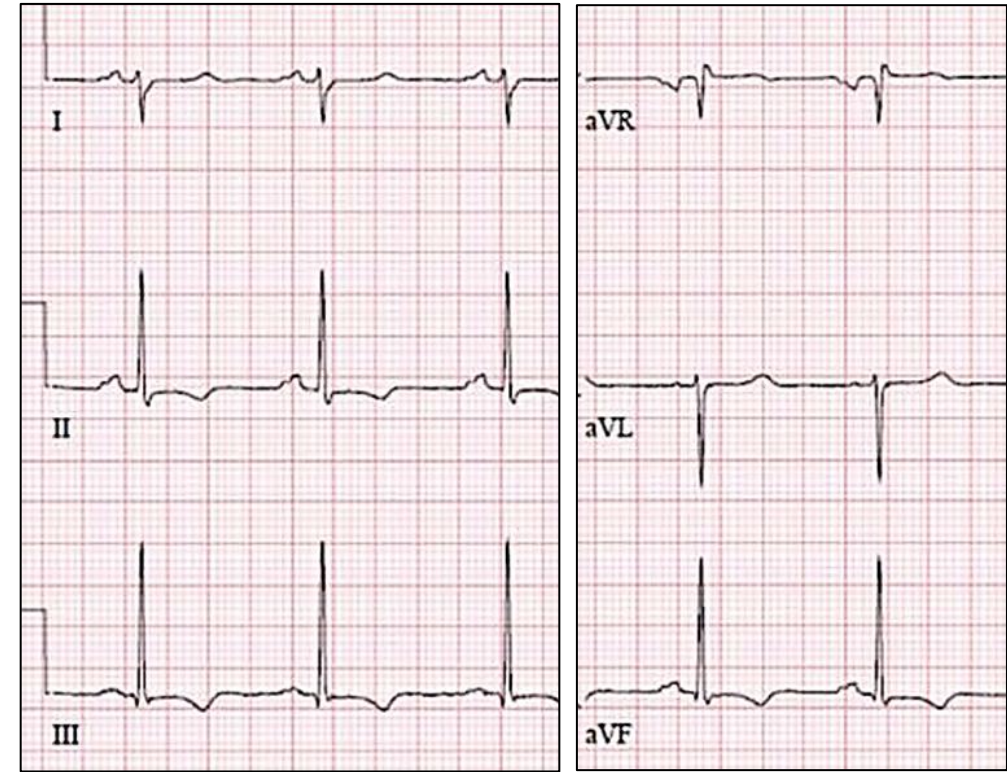
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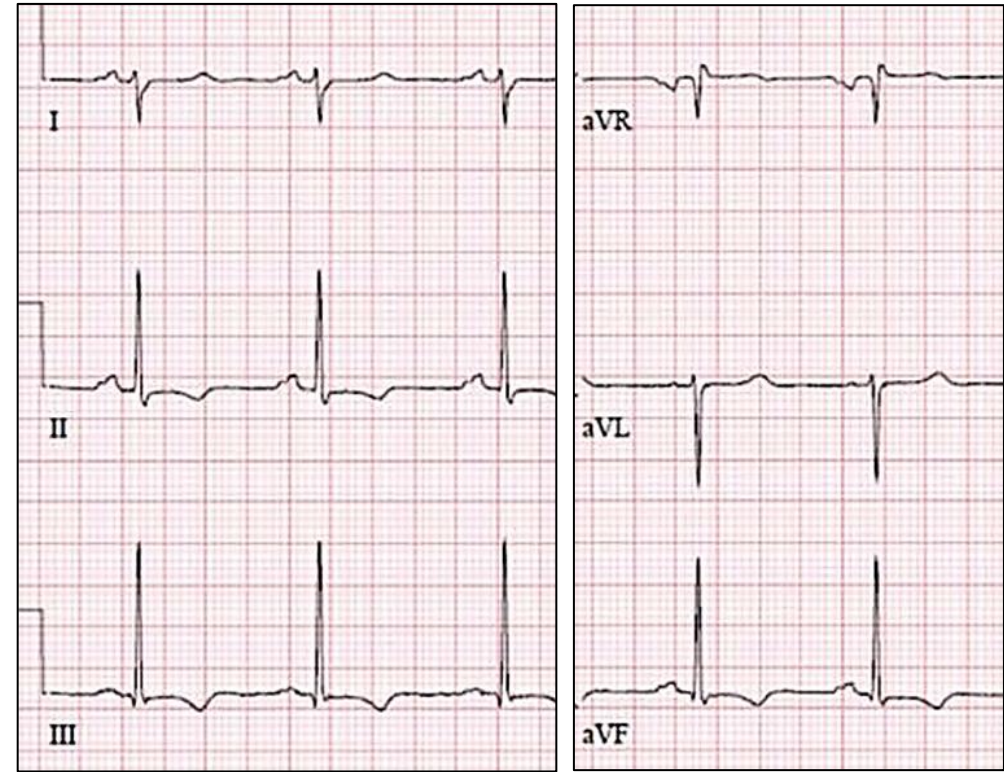
SOM.MK.II.BPM1.3.CPR.2.PHYS.0174 Explain the electrocardiographic changes associated with the myocardial ischemia, injury and cell death; and the concept of injury current and its relationship with the ST segment elevation or depression.

14. A 78-year-old man comes to the physician for a follow-up examination. He has a 20-year history of chronic obstructive pulmonary disease. Vital signs are within normal limits. Cardiac examination shows a loud S2 in the pulmonary area and a parasternal heave. An ECG in limb leads is shown. Which of the following findings regarding this patient's heart is most likely?

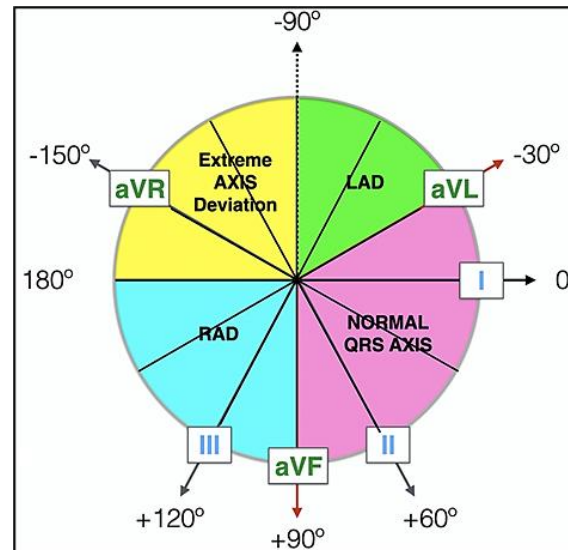


- A. Left axis deviation
- B. Left ventricular hypertrophy
- C. Normal axis of the heart
- D. Right axis deviation
- E. Extreme axis deviation

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15. A 71-year-old woman comes to the physician because of a 5-week history of dizziness and lightheadedness. She says she loses balance while getting off the chair. She has no chronic illnesses. She is taking no medications. Her blood pressure is 132/78 mm Hg and respirations are 16/min. Physical examination shows no abnormalities. Results of laboratory studies are within the reference ranges. An ECG recorded in lead II is shown. Which of the following electrical abnormalities is most likely in this patient's heart?



- A. Block in AV nodal conduction
- B. Abnormal atrial depolarization
- C. Abnormal ventricular depolarization
- D. Prolongation of AV nodal delay
- E. Reduced SA node excitability

A	B	C	D	E
0	0	0	0	0

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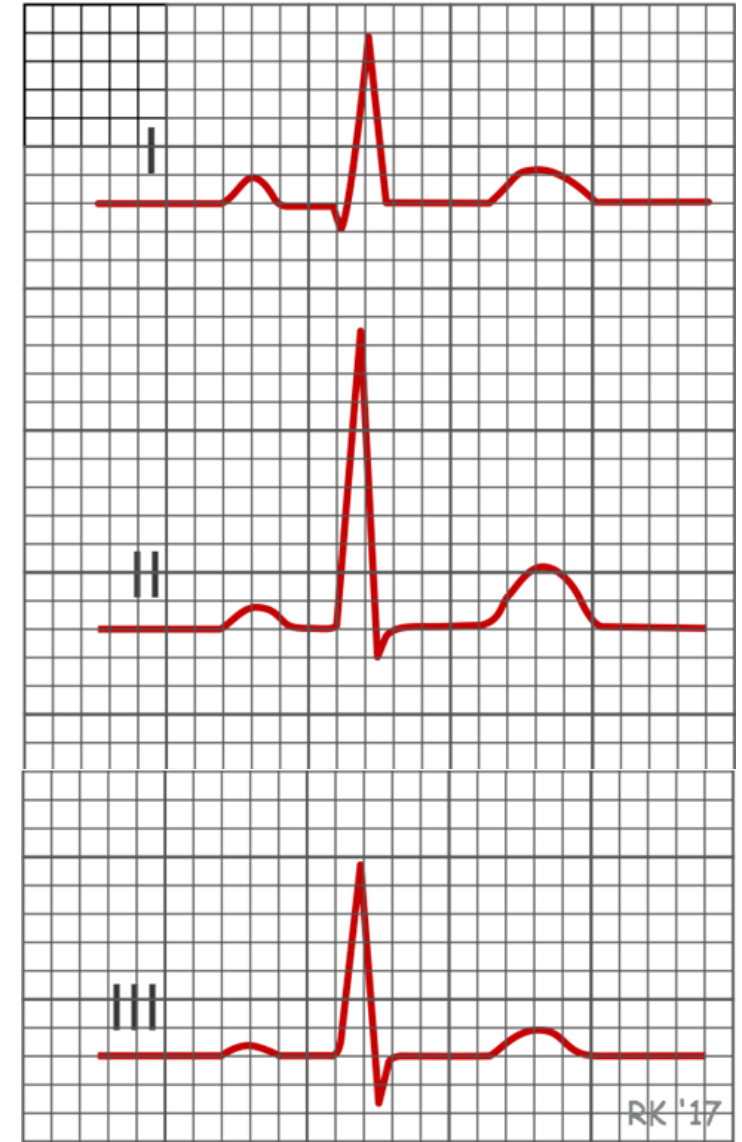


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- B. Abnormal atrial depolarization
- C. Abnormal ventricular depolarization
- D. Prolongation of AV nodal delay
- ✓ E. Reduced SA node excitability

SOM.MK.II.BPM1.3.CPR.2.PHYS.0178. Describe the alteration in conduction responsible for most common arrhythmias: tachycardia, bradycardia, A-V block, atrial and ventricular fibrillation

16. A 51-year-old man comes to the physician because of a 2-week history of shortness of breath while climbing stairs. She has a family history of hypertension. Her blood pressure is 128/86 mm Hg; other vital signs are within normal limits. Physical examination shows no abnormalities. An ECG in leads I and II are shown. Which of the following values best approximates the QRS complex in lead III in this patient? (Each small box=0.1 mV)

- A. 0.25 mV
- B. 0.30 mV
- C. 0.50 mV
- D. 0.65 mV
- E. 0.80 mV



0 0 0 0 0
A. B. C. D. E.



16. A 51-year-old man comes to the physician because of a 2-week history of shortness of breath while climbing stairs. She has a family history of hypertension. Her blood pressure is 128/86 mm Hg; other vital signs are within normal limits. Physical examination shows no abnormalities. An ECG in leads I and II are shown. Which of the following values best approximates the QRS complex in lead III in this patient? (Each small box=0.1 mV)

- A. 0.25 mV
- B. 0.30 mV
- ✓ C. 0.50 mV
- D. 0.65 mV
- E. 0.80 mV

$$\begin{aligned}\text{Lead II} &= \text{Lead I} + \text{Lead III} \\ \text{Lead III} &= \text{II} - \text{I} = 10 - 5 = 5 = 0.5 \text{ mV}\end{aligned}$$

SOM.MK.II.BPM1.3.CPR.2.PHYS.0172 Name the parts of a typical bipolar (Lead II) ECG tracing and explain the relationship between each of the waves, intervals, and segments in relation to the electrical state of the heart.

